

Programming for Economist

PS 6

Class 9

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Plan for today

1. Classes
2. Hints for PS6

1. Classes

- Classes are blueprints for creating **objects**
- They contain data accessed as **attributes** and class specific functions called **methods**.
- You can create multiple **instances** of a class with their own set of attributes and methods.
- Today you will see, how we through one class, can solve different economic models through the use of multiple instances.
- Today we start by working with a class used in the lectures. We later modify it.

1. Classes

We Define a Class by:

```
class ClassName(InheritedClasses*):
```

Initialise the class with:

```
def __init__(self, attributes*):
```

- 'self' represents the instance of the class
- Set attributes by `self.AttributeName = attribute`
- To inherit all attributes and methods from parent class use:

```
super().__init__(ParentClassAttributes*)
```

- Define one or more methods with:

```
def MethodName(self, args*):
```

- Create an instance of the class with

```
ObjectName = ClassName(attributes*):
```

- Access attributes and methods of the specific class instance

```
ObjectName.AttributeName
```

```
ObjectName.MethodName(args*)
```

2. Hints for PS6

- General
 - Keep in mind that running a new simulation for the same model object will override the previous simulation – save results somewhere!
- Q2
 - $f(\tilde{k}) = \tilde{k}^\alpha$
- Q3
 - Let the new class be a child class of the SolowModel class using the `super().__init__` constructor. Then you only need to set the new attributes and override the functions that have different in/outputs.
- Q5
 - To have savings rates drawn in logs, use the `log(mean_s)` as the mean when calling the `draw_ar1`, and then `np.exp('output')` to get the right savings rate.
 - When looping through economies store: capital per effective worker and the average growth rates in containers with shape (N,T) and (N)